Lakshya JEE 2.0 (2024)

Differential Equation

DPP-01

1. The order and degree of the differential equation

$$r=\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$ are respectively

(1) 2, 2 (2) 2, 3
(3) 2, 1 (4) None of these

2. If p and q are order and degree of differential equation

$$y^2\left(\frac{d^2y}{dx^2}\right)^2+3x\left(\frac{dy}{dx}\right)^{1/3}+x^2y^2=\sin x$$, then:

(1) $p>q$ (2) $\frac{p}{q}=\frac{1}{2}$
(3) $p=q$ (4) $p<q$

3. The order of the differential equation whose general solution is given by $y=(C_1+C_2)\cos(x+C_3)-C_4e^{x+C}5$ where C_1, C_2, C_3, C_4, C_5 are arbitrary constant is

(1) 5 (2) 4
(3) 3 (4) 2

4. Differential equation whose general solution is $y=c_1x+c_2/x$ for all values of c_1 and c_2 is

(1) $\frac{d^2y}{dx^2}+\frac{x^2}{y}+\frac{dy}{dx}=0$

(2) $\frac{d^2y}{dx^2}+\frac{y}{x^2}-\frac{dy}{dx}=0$

(3) $\frac{d^2y}{dx^2}+\frac{1}{2x}\frac{dy}{dx}=0$

(4) $\frac{d^2y}{dx^2}+\frac{1}{x}\frac{dy}{dx}-\frac{y}{x^2}=0$

5. The differential equation of all parabola having their axis of symmetry coinciding with the x-axis is

(1) $y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2=0$

(2) $y\frac{d^2x}{dy^2}+\left(\frac{dx}{dy}\right)^2=0$

(3) $y\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

(4) None of these

6. Which of the following is not a solution of $x^2y_1^2+xy\,y_1-6y^2=0$?

(1) $y=Cx^2$

(2) $x^2y=C$

(3) $\frac{1}{2}ln\,y=C+\log x$

(4) $x^3y=C$

7. The order and degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^{2/3}=4\frac{d^3y}{dx^3}$ are

(1) $\left(1,\frac{2}{3}\right)$ (2) (3, 1)
(3) (3, 3) (4) (1, 2)

8. Family $y=Ax+A^3$ of curves will corresponding to a differential equation of order

(1) 3 (2) 2
(3) 1 (4) Not defined

9. If $y=e^{(k+1)x}$ is a solution of differential equation $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+4y=0$, then $k=$

(1) −1 (2) 0
(3) 1 (4) 2

10. The differential equation whose solution is $Ax^2 + By^2 = 1$, where A and B are arbitrary constants is of

(1) second order and second degree

(2) first order and second degree

(3) first order and first degree

(4) second order and first degree

11. The order of the differential equation whose solution is $y = a\cos x + b\sin x + ce^{-x}$, is

(1) 3 (2) 1

(3) 2 (4) 4

12. The differential equation obtained by eliminating the arbitrary constants a and b from $xy = ae^x + be^{-x}$ is

(1) $x\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} - xy = 0$

(2) $\dfrac{d^2y}{dx^2} + 2y\dfrac{dy}{dx} - xy = 0$

(3) $x\dfrac{d^2y}{dx^2} + 2\dfrac{dy}{dx} + xy = 0$

(4) $\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} - 2y = 0$

13. Solution of $\dfrac{dy}{dx} = 3^{x+y}$ is

(1) $3^{x+y} = C$ (2) $3^x + 3^y = C$

(3) $3^{x-y} = C$ (4) $3^x + 3^{-y} = C$

14. Order and degree of the differential equation

$$\frac{d^2y}{dx^2} = \left\{y + \left(\frac{dy}{dx}\right)^2\right\}^{1/4}$$ are

(1) 4 and 2 (2) 1 and 2

(3) 1 and 4 (4) 2 and 4

15. If m and n are the order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^5 + 4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\frac{d^3y}{dx^3}} + \frac{d^3y}{dx^3} = x^2 - 1,$$ then

(1) $m = 3, n = 3$

(2) $m = 3, n = 2$

(3) $m = 3, n = 5$

(4) $m = 3, n = 1$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (4)
3. (3)
4. (4)
5. (1)
6. (2)
7. (3)
8. (3)
9. (3)
10. (4)
11. (1)
12. (1)
13. (4)
14. (4)
15. (2)

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